

Supplementary Material (Online Resource 1)

Real-Time Multi-Behavior Detection of Group-Housed Pigs on Edge Devices: Class-Imbalance-Aware Sampling and a Lightweight FasterNet Backbone — Supplementary Material

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Table S1. Per-class AP50 on the validation set. On the held-out test set, the key rare-class result is active 0.526 → 0.639 (baseline → M4, +11.3 points); sitting 0.534 → 0.571; fight 0.840 → 0.871.

Class	Baseline	M3 (FasterNet)	M4 (sampling)	M5 (combined)	M4 – baseline
active	0.459	0.372	0.552	0.452	+9.3
drink	0.408	0.440	0.459	0.430	+5.1
eat	0.436	0.437	0.447	0.449	+1.1
fight	0.858	0.828	0.842	0.786	−1.6
investigating	0.581	0.571	0.570	0.577	−1.1
lying	0.783	0.764	0.766	0.743	−1.7
nose-to-nose	0.686	0.655	0.622	0.710	−6.4
sitting	0.403	0.233	0.423	0.349	+2.0
standing	0.438	0.429	0.449	0.445	+1.1
walk	0.687	0.646	0.686	0.667	−0.1
mAP50 (all)	0.573	0.537	0.582	0.561	+0.9

Table S2. Per-class AP50 on the held-out test set for the baseline and the two adopted variants. The rare-class pattern mirrors the validation set (Table S1): the gains of M4 concentrate on active, sitting and fight; M5 keeps part of the rare-class gain while carrying 4.4% fewer parameters. All entries are single-run values (seed 0); the three-seed means for the overall metric are reported in Section 4.5 of the main text. M5 values were re-measured under the official validation protocol (overall mAP50 0.5933, matching the cloud-reported 0.5932).

Class	Baseline	M4 (sampling)	M5 (combined)
active	0.526	0.639	0.533
drink	0.389	0.373	0.408
eat	0.445	0.418	0.429
fight	0.840	0.871	0.849
investigating	0.552	0.562	0.588
lying	0.741	0.737	0.722
nose-to-nose	0.783	0.740	0.749
sitting	0.534	0.571	0.568
standing	0.487	0.468	0.475
walk	0.667	0.655	0.612

mAP50 (all)	0.596	0.604	0.593
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